6 Discussion

[To be completed: Discussion outline. (1) How much of measured connectedness is a common factor? About 41% of Diebold–Yilmaz connectedness disappears without the factor (range 38–43% across robustness checks); one factor explains 58% of the variance. What does this say about contagion versus common exposure (Forbes and Rigobon, 2002; Rey, 2013)? Why do the common share and DY connectedness move together, and why is that partly mechanical? (2) Does the method change the picture? Lasso and DY almost coincide, so regularization is not what matters; NETS keeps the pattern at a lower level; FNETS measures a different object (idiosyncratic part). Which method would you use for which purpose (risk monitoring, contagion analysis, portfolio diversification)? (3) Transmitters and receivers: South Africa, Korea, Taiwan versus Turkey, Vietnam, Argentina; how much may reflect trading hours of US-listed funds; what changes once the factor is removed. (4) Limitations: returns instead of volatilities; ETF prices embed exchange rates and US trading hours; short windows for NETS and FNETS (positive-definite adjustment, empty idiosyncratic network in the full sample, tuning-driven drops); almost no lagged links survive in the sparse methods.]

7 Conclusion

[To be completed: Conclusion outline. Answer the research question in three or four sentences. What can be claimed with confidence: weekly connectedness among emerging markets is high and mostly contemporaneous; a large and stable share of it (about 40%) is a common factor; the dense methods agree with each other. What cannot be claimed: the exact direction of spillovers between countries, or the idiosyncratic network structure, given the short windows. One sentence on extensions (volatility connectedness, longer samples, more markets).]

8 References

Matteo Barigozzi and Christian Brownlees. NETS: Network estimation for time series. *Journal of Applied Econometrics*, 34(3):347–364, 2019.

Matteo Barigozzi, Haeran Cho, and Dom Owens. FNETS: Factor-adjusted network estimation and forecasting for high-dimensional time series. *Journal of Business & Economic Statistics*, 42(3):890–902, 2024.

T. Tony Cai, Weidong Liu, and Harrison H. Zhou. Estimating sparse precision matrix: Optimal rates of convergence and adaptive estimation. *The Annals of Statistics*, 44(2):455–488, 2016.

Jiahua Chen and Zehua Chen. Extended Bayesian information criteria for model selection with large model spaces. *Biometrika*, 95(3):759–771, 2008.

Francis X. Diebold and Kamil Yılmaz. On the network topology of variance decompositions: Measuring the connectedness of financial firms. *Journal of Econometrics*, 182(1):119–134, 2014.

Kristin J. Forbes and Roberto Rigobon. No contagion, only interdependence: Measuring stock market comovements. *The Journal of Finance*, 57(5):2223–2261, 2002.

Mario Forni, Marc Hallin, Marco Lippi, and Lucrezia Reichlin. The generalized dynamic-factor model: Identification and estimation. *Review of Economics and Statistics*, 82(4):540–554, 2000.

Marc Hallin and Roman Liška. Determining the number of factors in the general dynamic factor model. *Journal of the American Statistical Association*, 102(478):603–617, 2007.